Kumar Vanshaj

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AI engineer and CS undergraduate with hands-on experience in **agentic AI systems**, **LLM pipelines**, **prompt engineering**, and **full-stack development**

Experience

TheWordofmouth **Remote**

Software Developer Intern *Jan 2026 – Present*

- Engineered automated Python scrapers ingesting customer reviews and product images from Amazon and Myntra.
- Built **fault-tolerant data pipelines** on AWS S3 + MongoDB with structured/unstructured syncing; wrote **PyTest integration tests** covering core pipeline flows.
- Deployed background worker services on **AWS EC2** for continuous processing; collaborated with a 2-person backend team on database schema design and query optimization.
- Built a complete **Shopify app** managing multiple stores, integrating custom API endpoints with **JWT-based auth** and synchronizing state with the company's core backend.

YesCity **Remote**

Full-Stack Developer Intern *Oct 2025 – Dec 2025*

- Optimized 12+ RESTful API endpoints serving **500+ daily active users**, cutting average response time by **35%** through query rewriting and server-side caching.
- Built the complete backend of an advertisement system with **graph-based data modeling**, Razorpay payment processing, and **JWT auth with role-based access control (RBAC)**.
- Reduced chatbot response latency by **40%** by consolidating API calls and eliminating redundant DB round-trips; worked within a cross-functional team of 3 engineers.

Freelance **Remote**

Full-Stack Developer *May 2025 – July 2025*

- Designed and deployed **Check Meals**, a QR-based restaurant ordering platform handling **100+ concurrent sessions** with live menu updates and cross-device responsiveness.
- Reduced deployment time from 15 min to under 2 min by containerizing with Docker and automating CI/CD via GitHub Actions.

Projects

VectorMindsAI | *Next.js, TypeScript, LangChain, Inngest, Pinecone, Groq*

- Engineered a **multi-agent orchestration system** with 4 specialized agents (Planner, Researcher, Reviewer, Prompt Enhancer) for automated research workflows.
- Designed **structured prompt workflows** with iterative LLM evaluation; built **RAG pipelines** using dense **embedding models** and Pinecone for semantic retrieval across 5 LLM providers with automatic fallback.
- Wired 8 external services (Tavily, Pinecone, Sentry, PostHog) and containerized with **Docker multi-stage builds**, cutting image size by 40%.

LinkedIn Automation Agent | *Python, LangChain, Playwright, Groq*

- Built an **end-to-end browser automation agent** that autonomously navigates LinkedIn, extracts leads, and drafts personalized outreach reducing manual effort by **80%**.
- Designed a **tool-use pipeline** with Groq-powered LLM inference for real-time decision making: selecting actions, filling forms, and handling dynamic page states.
- Implemented **rate-limiting, session management, and cookie persistence** to operate reliably within platform constraints across long-running sessions.

PriceIQ | *Next.js, Node.js, MongoDB, Telegram API*

- Built a **price-tracking dashboard** monitoring **100+ products** across e-commerce platforms with historical trend visualizations.
- Automated pipelines running every 6 hours with **request throttling** (max 10 req/min) and an event-driven **Telegram notification service** for instant price-drop alerts.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, Bash, SQL
Frameworks & Libraries: React.js, Next.js, Node.js, FastAPI, Django, Tailwind CSS, LangChain, Playwright
AI & Agentic Systems: Multi-agent Orchestration, RAG Pipelines, Prompt Engineering, Embedding Models, Transformer Architecture, Vector Databases (Pinecone), LLM Tool-use, Fine-tuning (LoRA/PEFT), Claude Code, Groq, Inngest, Tavily
Backend & Security: RESTful APIs, JWT Auth, RBAC, Session Management, Graph Data Modeling, Shopify API, Razorpay, PostHog
DevOps & Cloud: Docker, Docker Compose, GitHub Actions, AWS (EC2, S3, Worker Services)
Testing & Tools: PyTest, Postman, Selenium, Linux, Git
CS Fundamentals: Data Structures & Algorithms, System Design, OOP, Computer Networks, Operating Systems, DBMS

Achievements

Open Source

- Contributed **5+ merged PRs** to open-source repositories including Apache, Wagtail, and JSON Schema.

Competitive Programming

- **LeetCode**: 500+ problems solved (rating: 1700+) | **CodeChef**: 3-star (rating: 1700+)

Leadership

- **Perplexity AI Campus Ambassador**: Organized AI workshops on campus and drove industry–academia engagement initiatives.

Education

IIIT Bhubaneswar **CGPA: 8.11 / 10**

Bachelor of Technology in Computer Science and Engineering *Bhubaneswar, Odisha | Expected May 2027*

- **Relevant Coursework**: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, OOP, Discrete Mathematics, Software Engineering